

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 2

MH1301 Discrete Mathematics

Tutorial 1

For the tutorial in Week 2 on 25 January, let us discuss

- Ex. 6.1.3, 6.1.4, 6.1.6, 6.1.7, 6.1.9, 6.1.10, 6.1.13, 6.1.13, 6.1.20, 6.1.21.

Ex. 6.1.3 and Ex. 6.1.4.

- (i) Assume there are 26 letters, how many different three-letter initials can people have ?
- (ii) A committee is formed consisting of one representative from each of the 50 states in the United States, where the representative from a state is either the governor or one of the two senators from that state, How many ways are there to form this committee ?

Ex. 6.1.6 and 6.1.7.

- (i) How many bit strings with length not exceeding n , where n is a positive integer, consist entirely of 1's, not counting the empty string ?
- (ii) How many bit strings of length n , where n is a positive integer, start and end with 1's ?

Ex. 6.1.10. There are 4 choices $\{G, U, A, C\}$ for each element of an RNA sequence, how many 6-element RNA sequences

- (a) do not contain U ?
- (b) end with GU ?
- (c) start with C ?
- (d) contain only A or U ?

Ex. 6.1.12 (a)–(c). How many positive integers between 100 and 999 (both inclusive)

- (a) are divisible by 7 ?
- (b) are odd ?
- (c) have the same three decimal digits ?

Ex. 6.1.12 (d)–(h). How many positive integers between 100 and 999 (both inclusive)

- (d) are not divisible by 4 ?
- (e) are divisible by 3 or 4 ?
- (f) are not divisible by either 3 or 4 ?
- (g) are divisible by 3 and 4 ?
- (h) are divisible by 3 but not by 4 ?

Ex. 6.1.13. How many strings of three decimal digits

- (a) do not contain the same digit three times ?
- (b) begin with an odd digit ?
- (c) have exactly two digits that are 4's ?

Ex. 6.1.9. There are 128 different ASCII characters, how many strings of 5 ASCII characters contain the character '@' at least once ?

Solution: There are in total 128^5 length-5 strings without any restriction, and 127^5 length-5 strings containing no '@', hence the answer is $128^5 - 127^5 = 1321368961$. $\square$

Ex. 6.1.20 and 6.1.21. Consider functions that maps from the set $A = \{1, 2, \dots, n\}$ to the set $B = \{0, 1\}$, where n is a positive integer,

- (a) how many such functions are there ?
- (b) how many such functions are one-to-one?
- (c) how many such functions are there that assign 0 to both 1 and n ?
- (d) how many such functions are there that assign 1 to exactly one element in A ?